AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 24 · STUDENT LESSON

Prototype a Grounded Assistant

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 18: missing evidence needs an explicit fallback.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of capstone grounded assistant. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of capstone grounded assistant.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

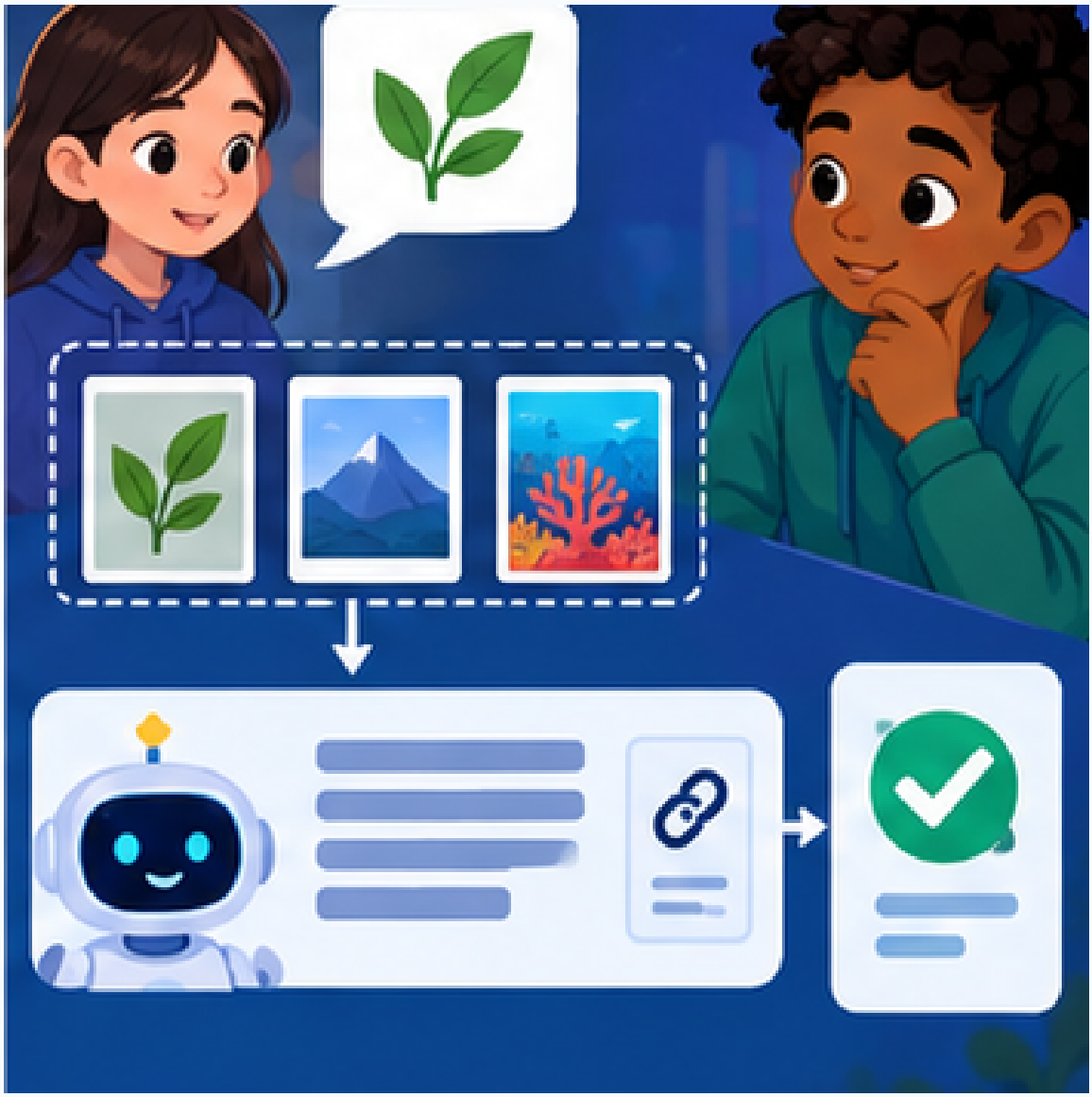
Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 24: **Prototype** a Grounded Assistant

A smallest working product connects prompt, references, interaction, and **output**.



CREATOR MISSION

Build one **end-to-end** Study Buddy path.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Capstone requirements and corpus. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can explain **prototype** a grounded assistant using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Prototype a Grounded Assistant case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **prototype** a grounded assistant to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Build the assistant as a sequence of testable stages: validate request, retrieve authorised **evidence**, decide answerability, draft, validate and return. Avoid merging all errors into one model score.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

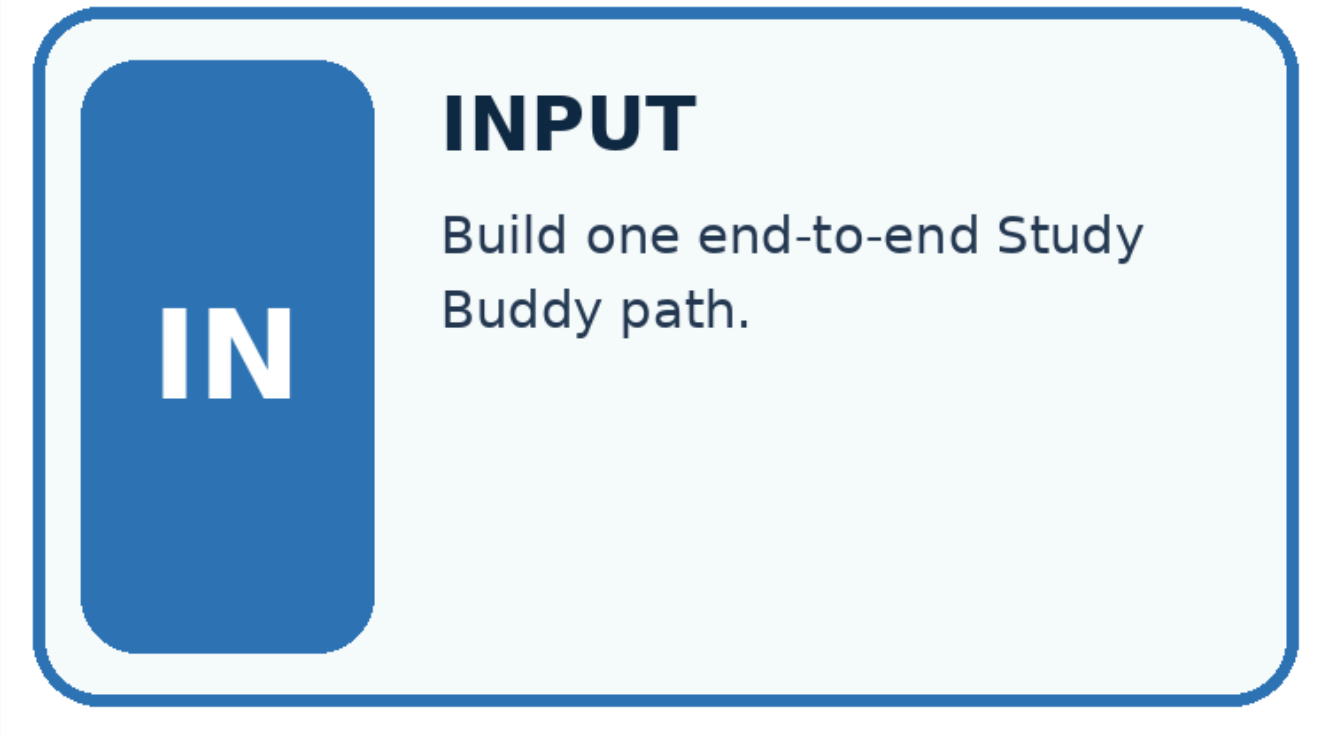
See how it works

> Follow the mechanism for Prototype a Grounded Assistant. Connect each labelled input, deciding step and result.

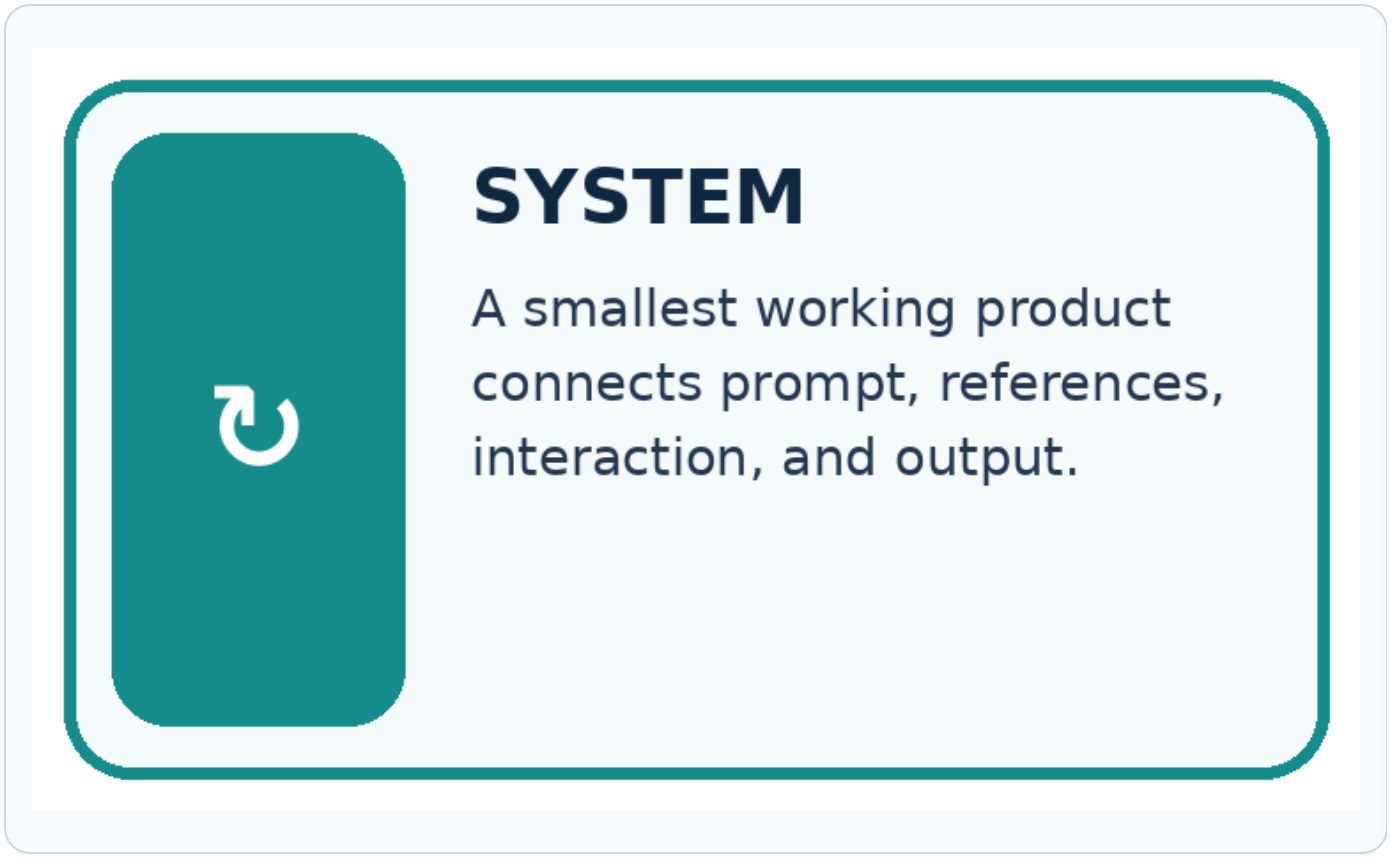
VISUAL MECHANISM

Input → Process → Output

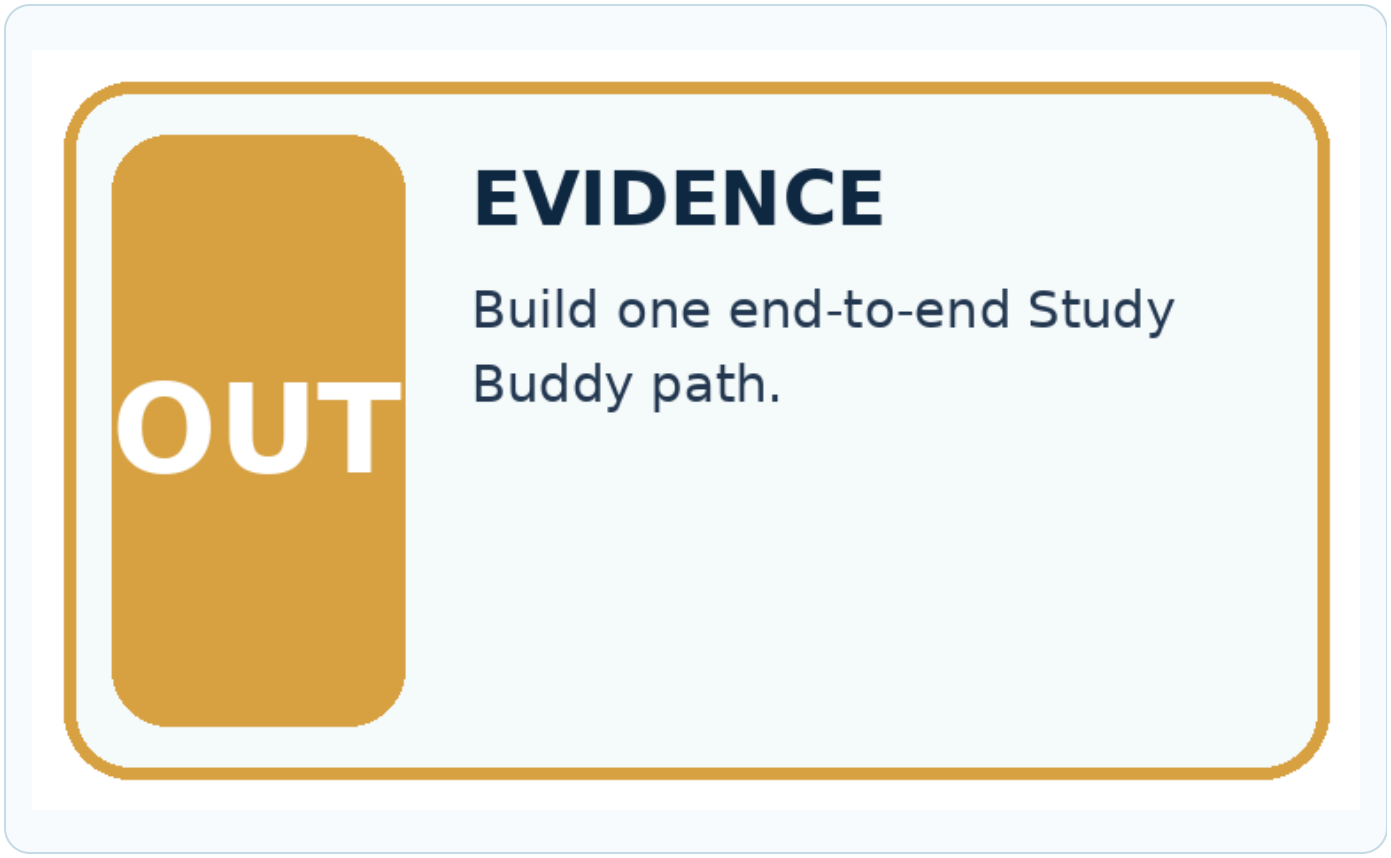
Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
SYSTEM
A smallest working product connects prompt, references, interaction, and output.

↓


OUT
EVIDENCE
Build one end-to-end Study Buddy path.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.

Capstone grounded assistant — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.	A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.	Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

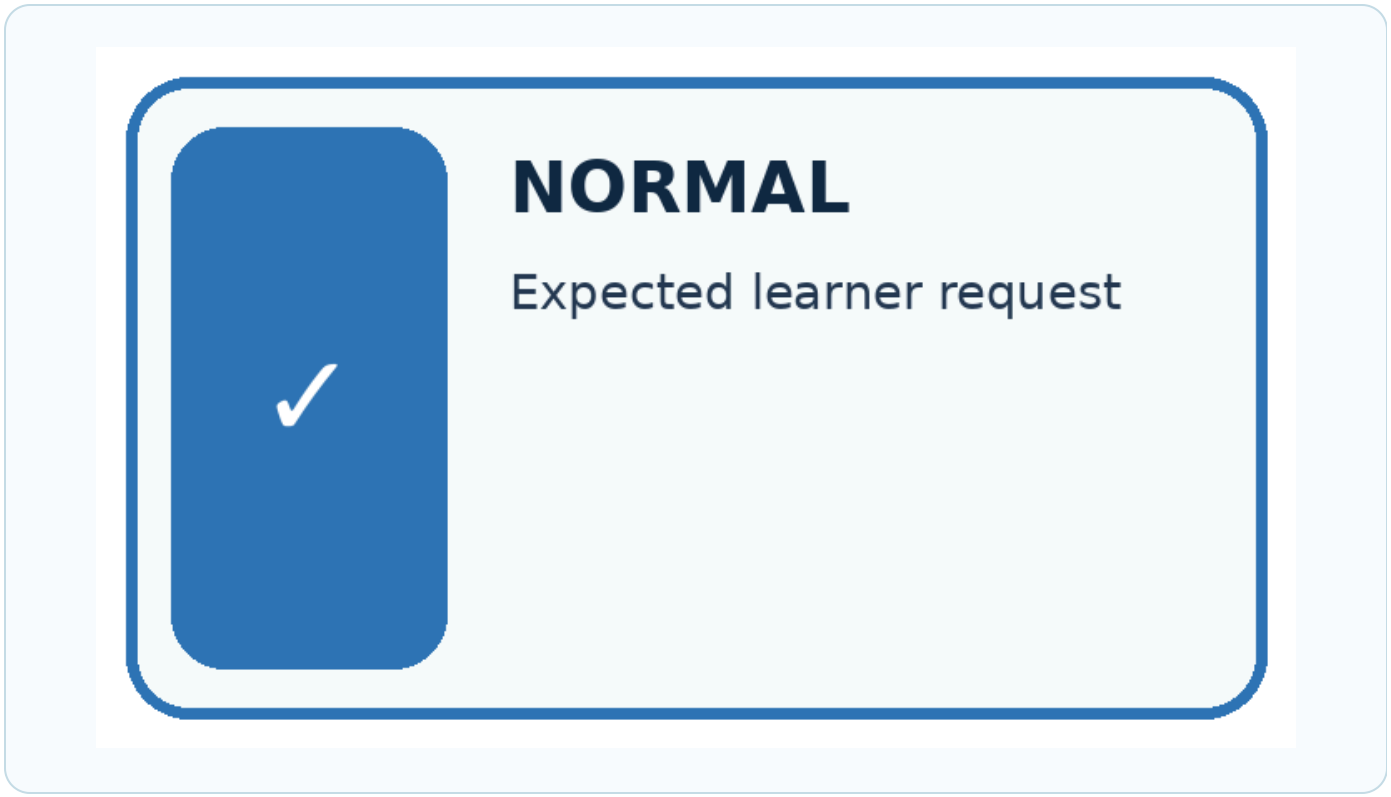
> Compare the supplied cases for Prototype a Grounded Assistant. Identify the detail that changes the conclusion.

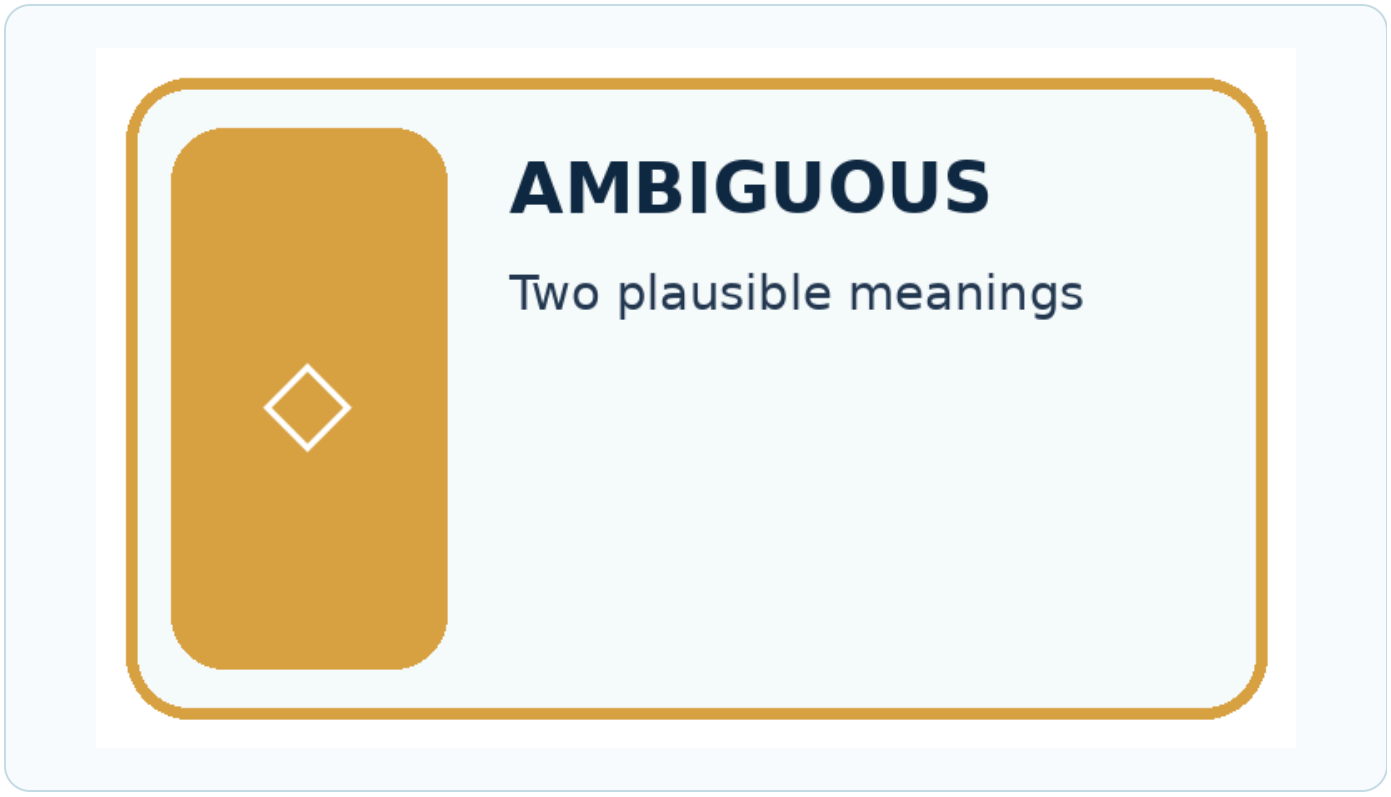
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: PL3 is absent while PL2 remains.

Predicted result

Prediction: Explain that the daily check is not specified in the available source.

Why the result follows

Reason: The remaining general water fact does not establish this activity's schedule.

STUDENT LESSON · 5 OF 18

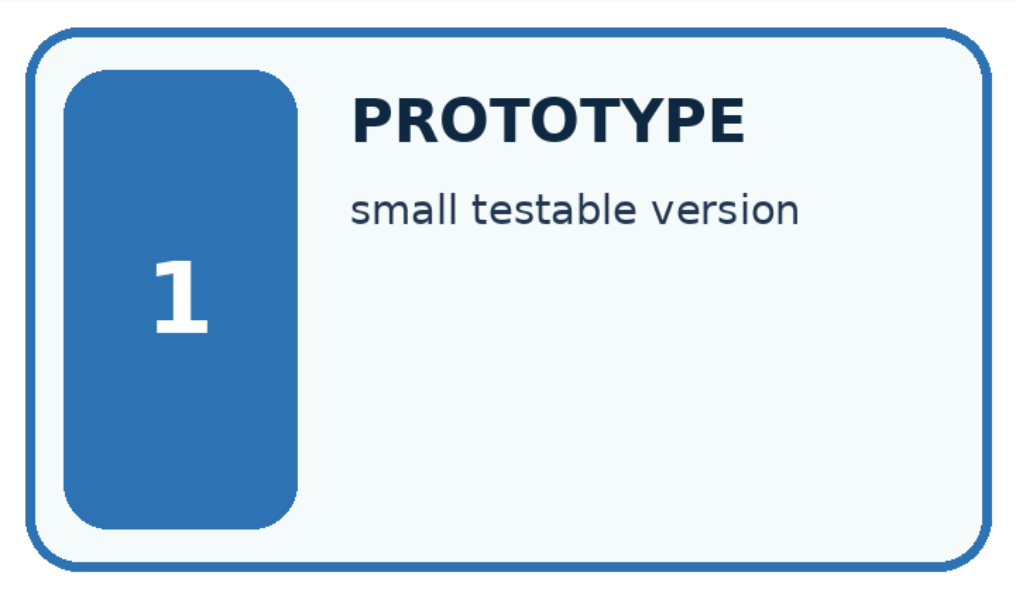
Words that unlock the idea

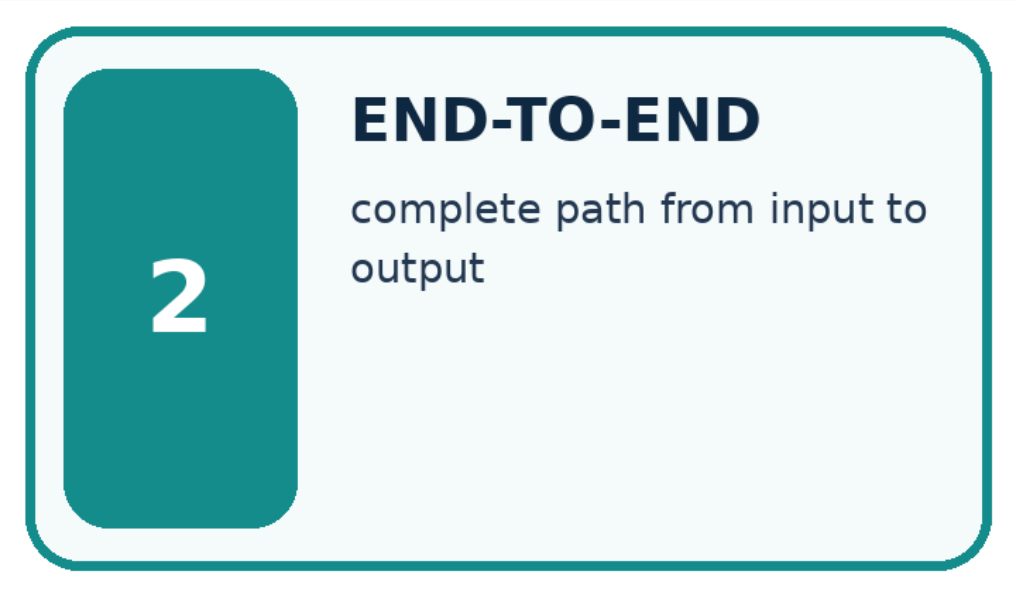
> Use the vocabulary of Prototype a Grounded Assistant to explain a concrete decision, rather than only memorising definitions.

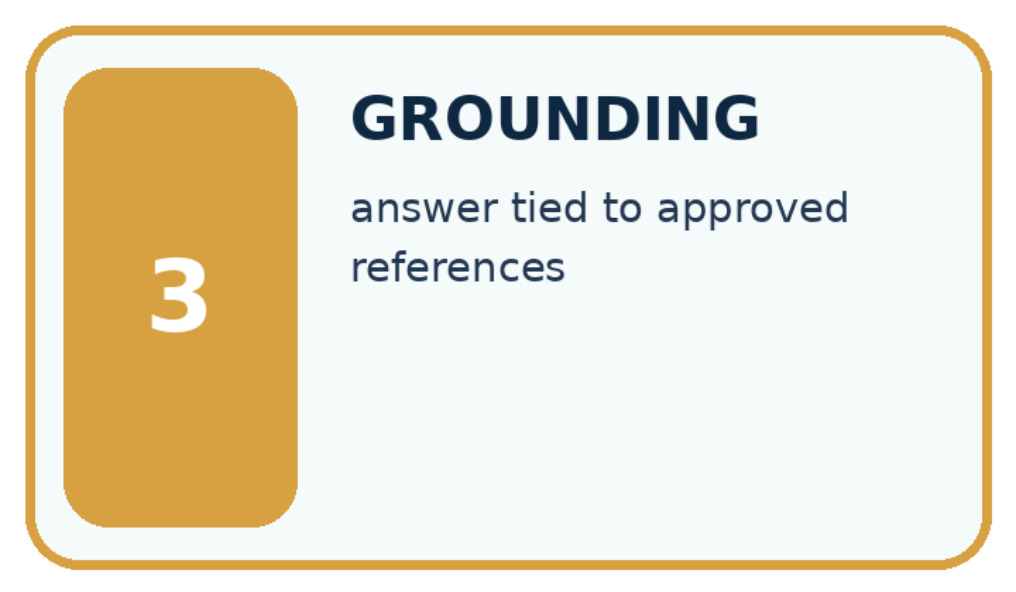
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
PROTOTYPE
small testable version


2
END-TO-END
complete path from input to output





Word	My own example	Not this / boundary
prototype	An early working or simulated version used to test assumptions about a design.	
end-to-end	Covering the entire process from original input to final output.	
grounding	Connecting an answer or action to specified evidence or observations.	

Word	My own example	Not this / boundary
fallback	A predefined alternative action used when the usual path cannot be completed reliably.	

Words used in the closer explanation

Word	Meaning
Pipeline stage	One operation with defined inputs and failure handling
Answerability gate	A check for sufficient evidence before answering
Semantic validator	A check that claims match task evidence

STUDENT LESSON · 6 OF 18

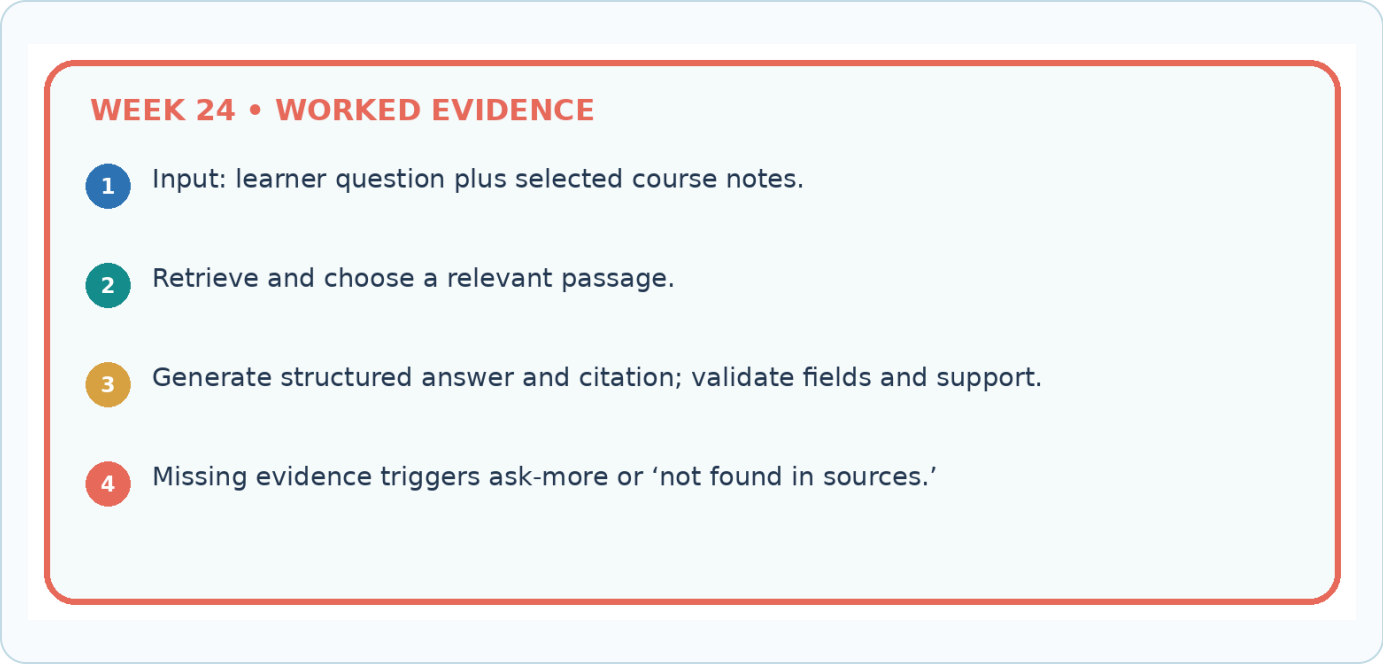
Follow a worked example

> Follow the worked trace for Prototype a Grounded Assistant, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Input**: learner question plus selected course notes.

2. Retrieve and choose a relevant passage.

3. Generate structured answer and citation; validate fields and support.

4. Missing **evidence** triggers ask-more or ‘not found in sources.’

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

Worked explanation · Prototype a Grounded Assistant

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prototype a Grounded Assistant

Build a minimal grounded study answer with checks between retrieval and display.

The facts we will use

Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/24

Prototype a Grounded Assistant

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve evidence

PL3 directly supports the daily soil-check claim; PL2 supplies a different fact.



2. Match question

Choose the activity's soil check, not a general watering recommendation.



3. Validate draft

Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

CHANGE ONE THING

PL3 is absent while PL2 remains.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve evidence	PL3 directly supports the daily soil-check claim; PL2 supplies a different fact.
2. Match question	Choose the activity's soil check, not a general watering recommendation.
3. Validate draft	Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A grounded **prototype** still needs an evaluation set and permission checks before anyone relies on it.

Keep the stated limit

Read the labelled picture: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: PL3 is absent while PL2 remains. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Prototype a Grounded Assistant claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A prototype is complete when the happy path works once. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

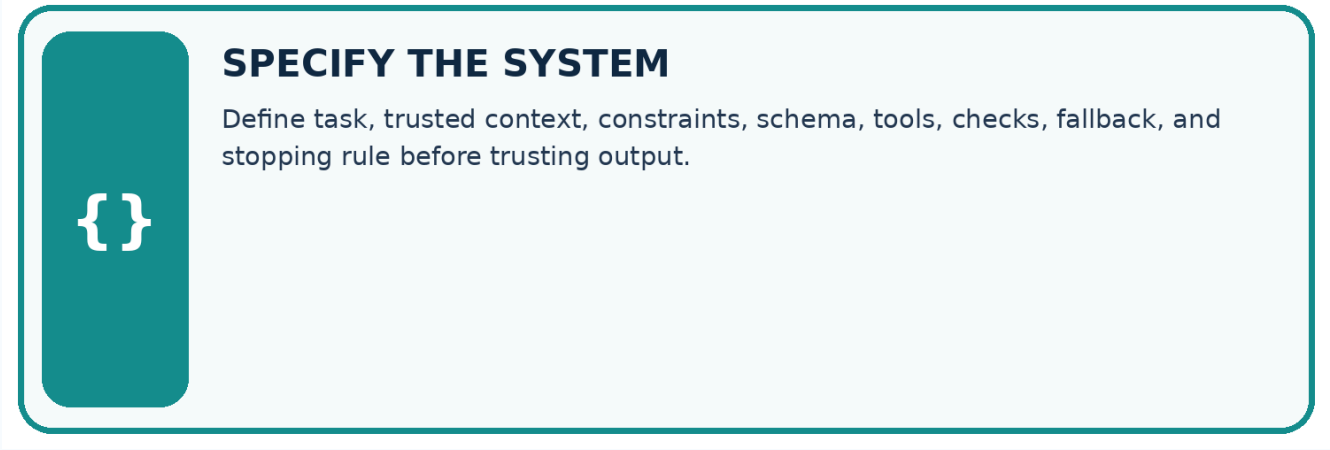
Practise together

> Practise Prototype a Grounded Assistant with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A correct happy path does not establish **fallback**, access control or side-effect safety.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone grounded assistant”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

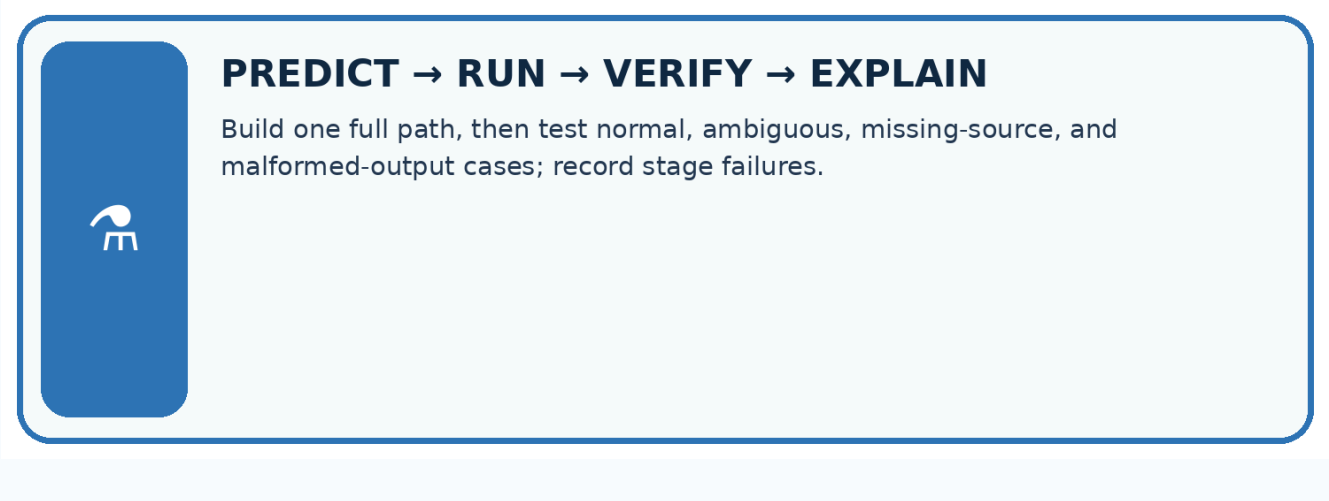
Run the investigation

> Investigate Prototype a Grounded Assistant. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Build one full path, then test normal, ambiguous, missing-source, and malformed-**output** cases; record stage failures.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

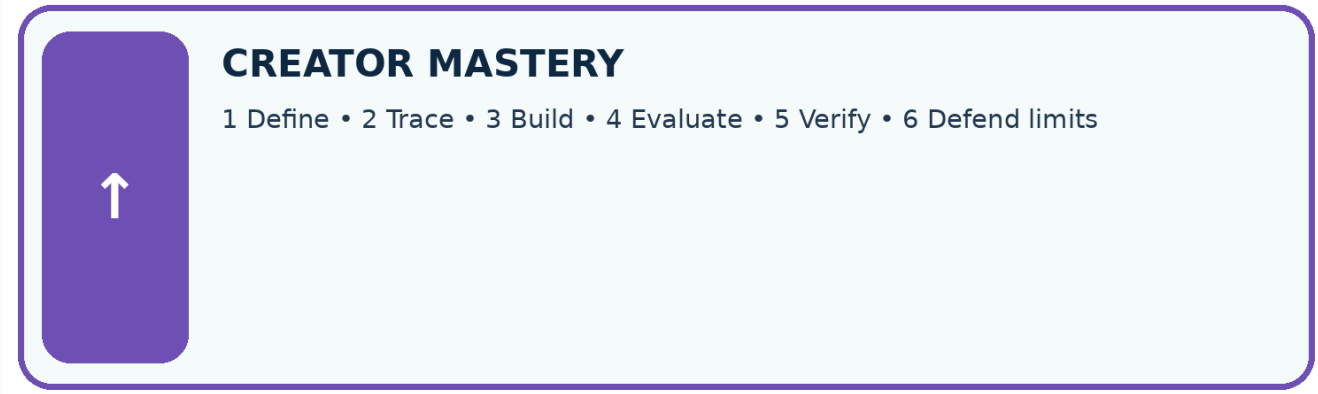
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Prototype a Grounded Assistant. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.

VERIFIED STUDY BUDDY

Connect user interface, source store, retrieval, prompt, schema validation, and evidence display.

THIS WEEK'S CHECKPOINT Connect user interface, source store, retrieval, prompt, schema validation, and **evidence** display.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

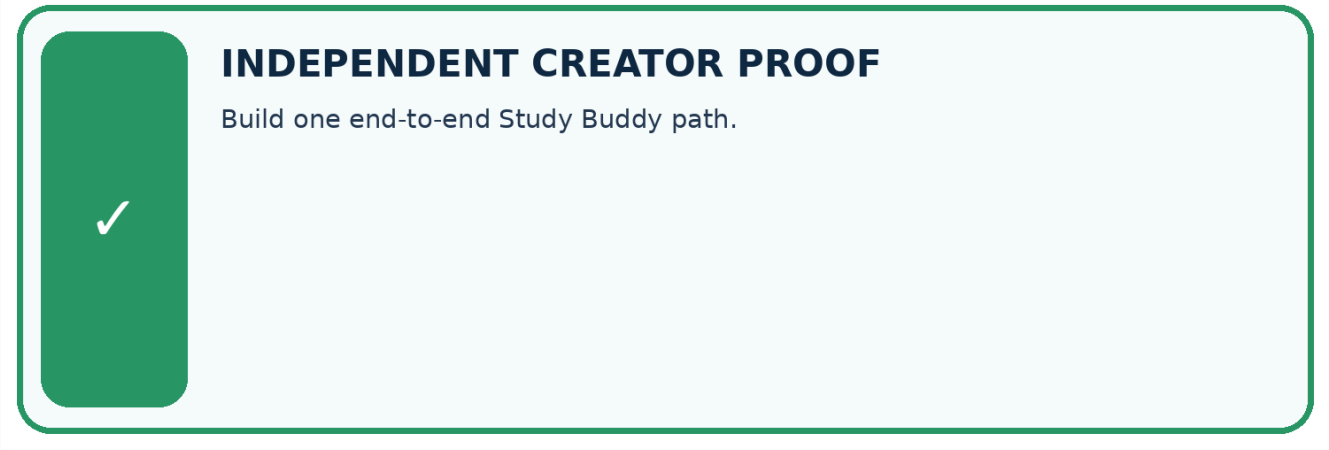
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Build one **end-to-end** Study Buddy path.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Prototype a Grounded Assistant case for supported practice. Repeated examples do not become fresh assessments.

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Approved notes say plants need light and roots take up water. Ask how roots help.

The question

Specify the minimum grounded path.

Worked explanation

Retrieve the roots note, answer that roots take up water and reference it. Validate support and avoid unrelated claims.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Build the smallest grounded path

Evidence or stage	Value or result
Question	How do roots help?
Retrieve	Roots take up water
Answer	Explain water uptake
Check	Reference and verify the roots note

Approved notes say plants need light and roots take up water. Ask how roots help.

Specify the minimum grounded path.

Retrieve the roots note, answer that roots take up water and reference it. Validate support and avoid unrelated claims.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A study assistant has a supplied passage about roots. A user asks a question about planets. What should the **prototype** do within a passage-only boundary?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Connect user interface, source store, retrieval, prompt, schema validation, and **evidence** display.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or **fallback**.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Prototype a Grounded Assistant, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Build a minimal grounded study answer with checks between retrieval and display.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone grounded assistant	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Prototype a Grounded Assistant without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone grounded assistant

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a minimal grounded study answer with checks between retrieval and display.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule.

1. Retrieve **evidence**

2. Match question

3. Validate draft

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Validate draft Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Read the picture in order.

Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: PL3 is absent while PL2 remains. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Explain that the daily check is not specified in the available source.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Explain that the daily check is not specified in the available source. The remaining general water fact does not establish this activity's schedule.

Repair a missing building block

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Tools and APIs (ai-3/21). Goal: Explain the **mechanism** and terminology of tools and api contracts. Short practice: Does a valid C7 ID permit the extra lending action?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

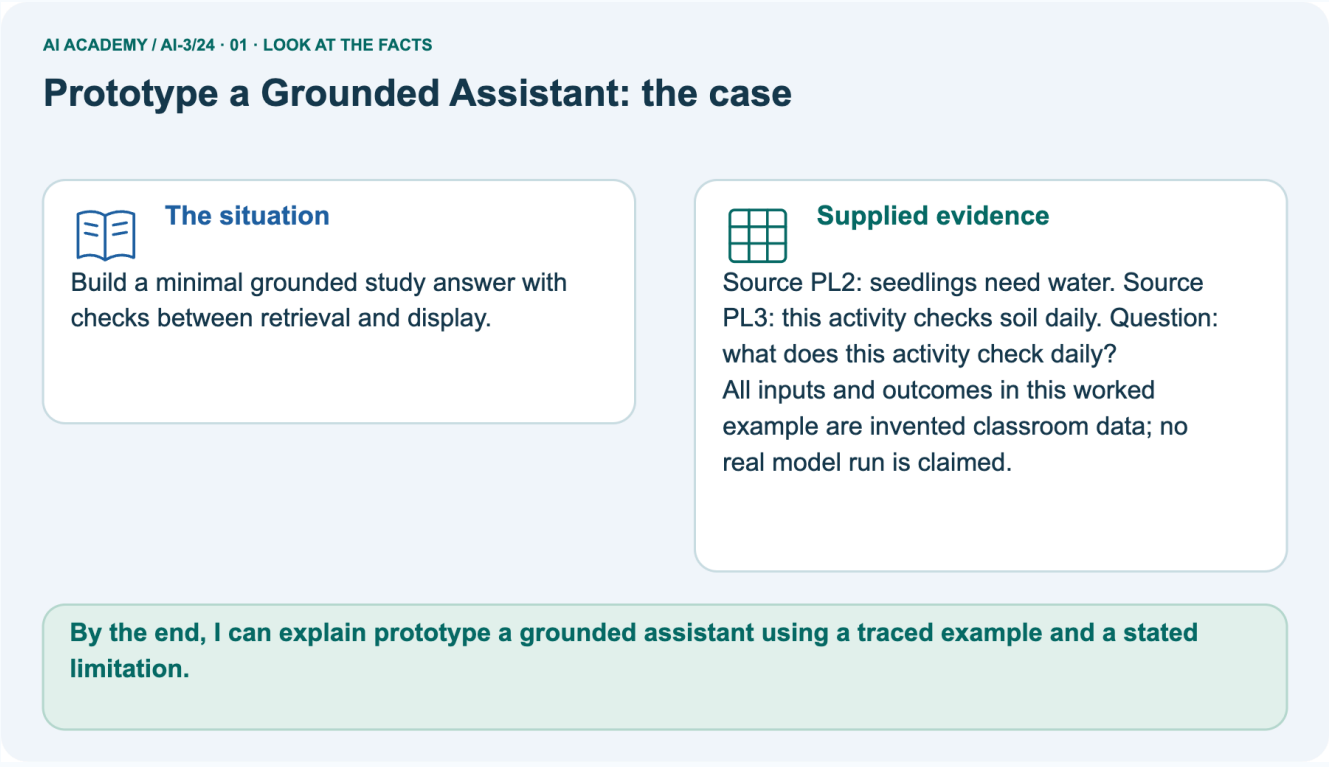
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Prototype a Grounded Assistant graphic using its labels, deciding evidence and relevant limitation.

01. Prototype a Grounded Assistant: the case



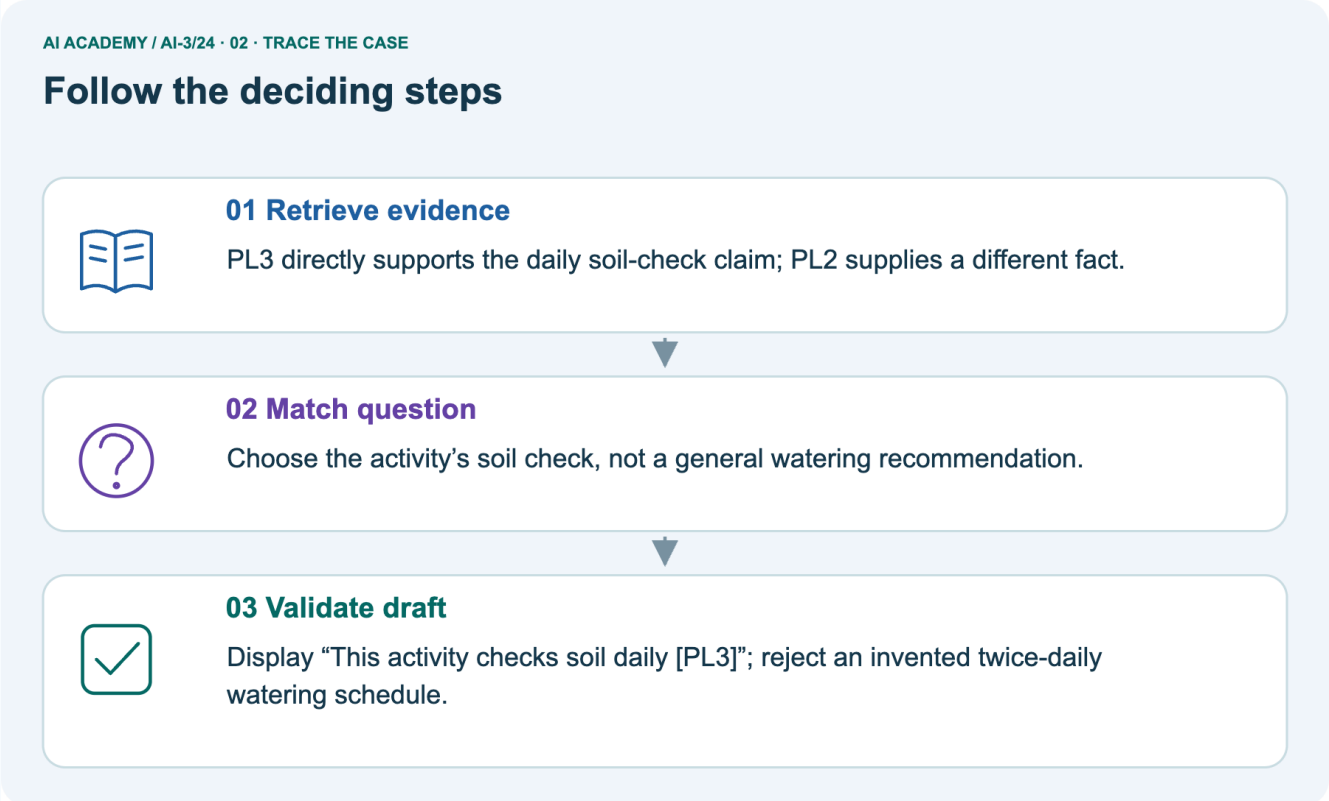
Follow the labelled picture with the matching explanation. Open the vector diagram

Build a minimal grounded study answer with checks between retrieval and display. Supplied facts: Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve **evidence**: PL3 directly supports the daily soil-check claim; PL2 supplies a different fact. → Match question: Choose the activity's soil check, not a general watering recommendation. → Validate draft: Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone grounded assistant — evidence trace

AI ACADEMY / AI-3/24 · 03 · INSPECT THE LESSON MODEL

Capstone grounded assistant — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.	A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.	Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

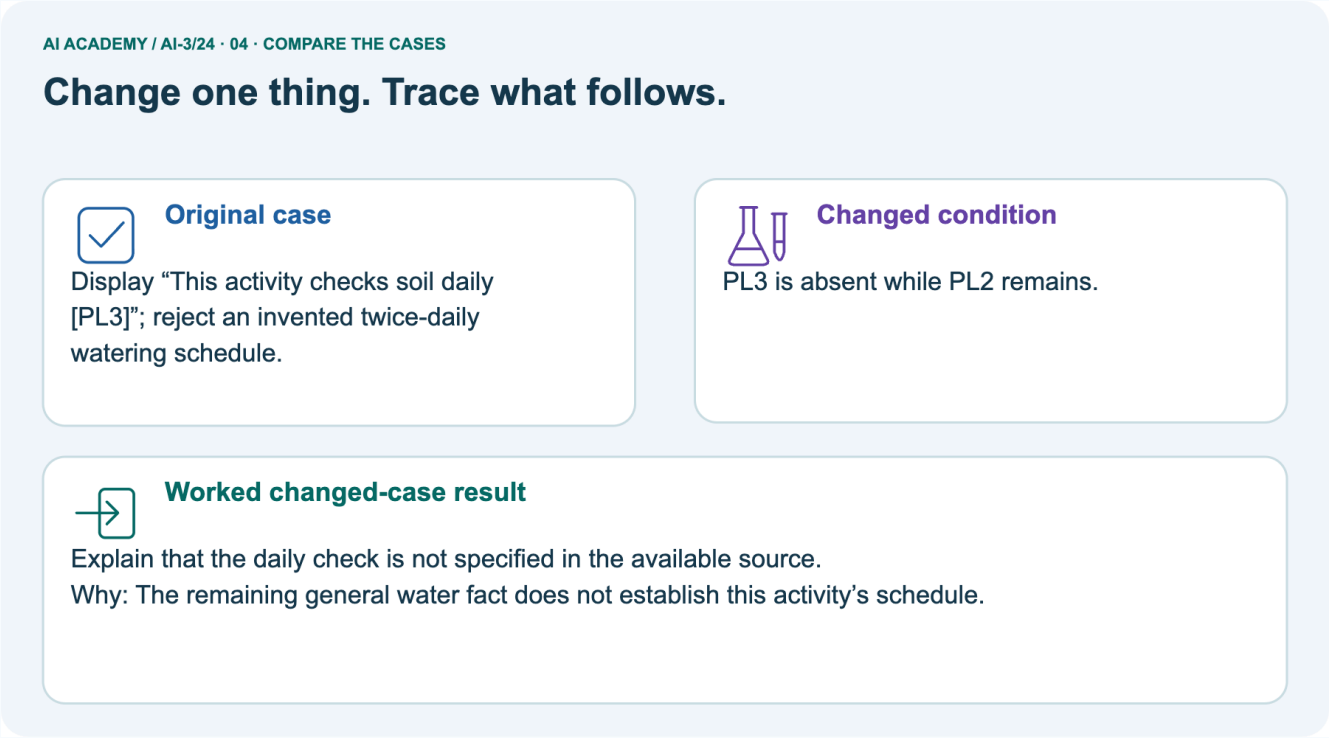
Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



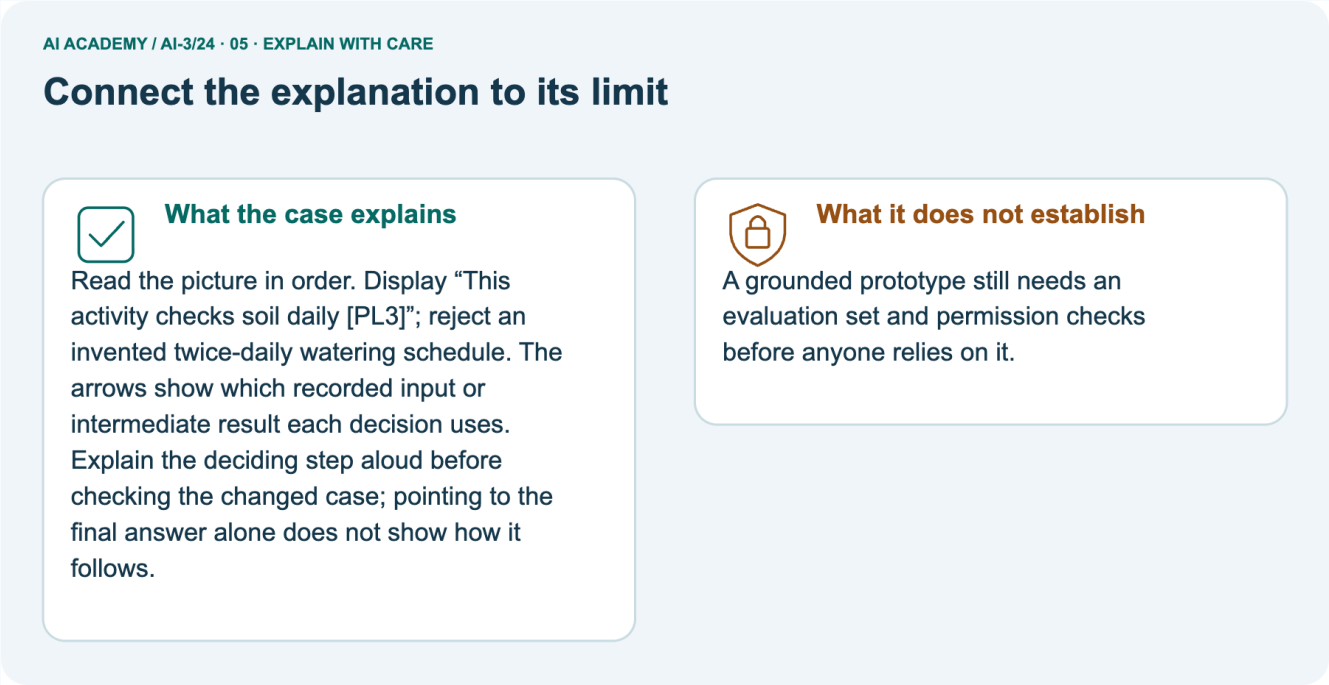
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Changed condition: PL3 is absent while PL2 remains.

Expected result: Explain that the daily check is not specified in the available source. Reason: The remaining general water fact does not establish this activity’s schedule.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

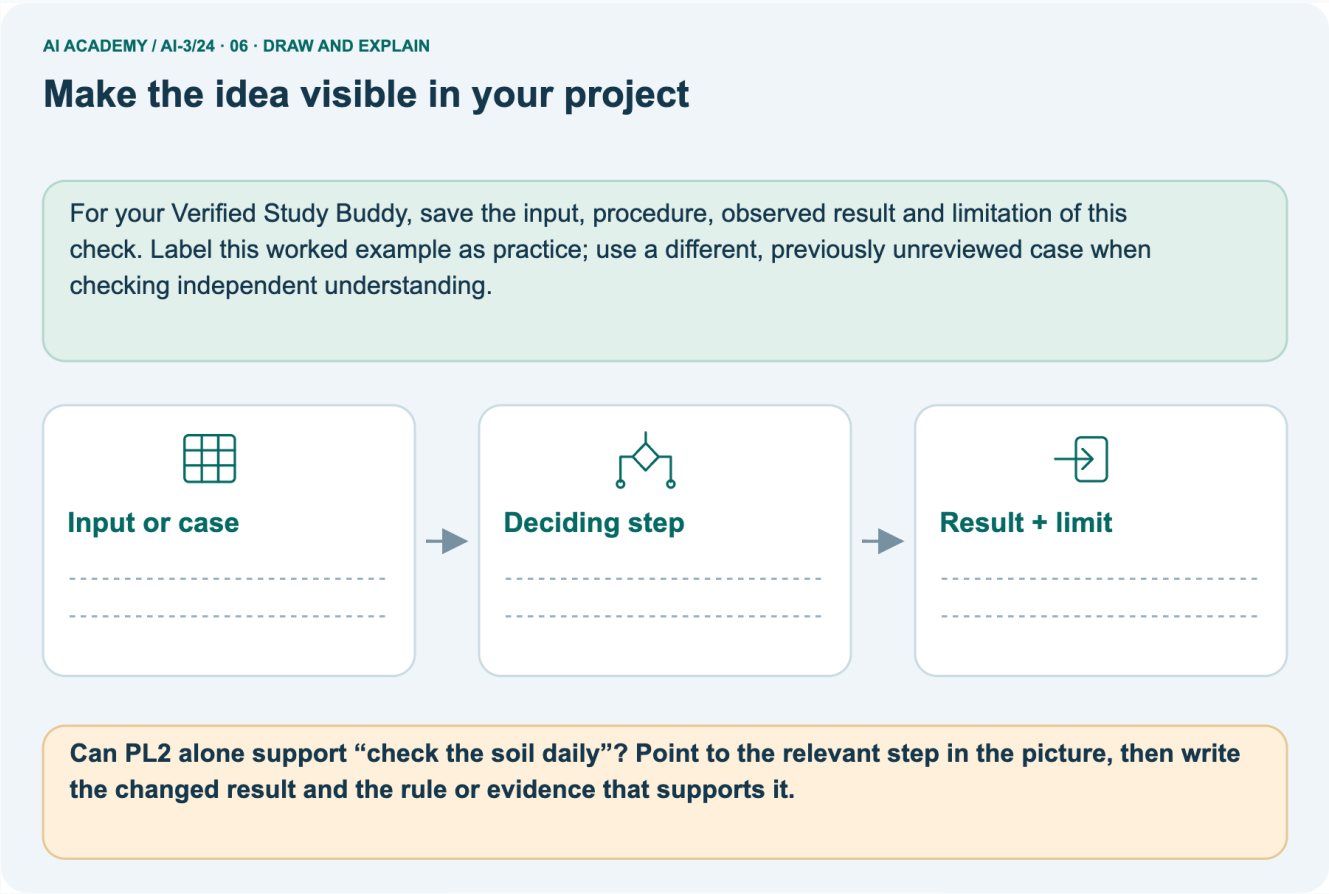
Read the picture in order. Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A grounded **prototype** still needs an evaluation set and permission checks before anyone relies on it.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.